

Hacker Challenge Guide

Three sponsor tracks. Six challenges. Pick one direction — dedicated or open — and decide for yourself what a great solution looks like.

Each dedicated challenge (a) is a focused problem set by the track sponsor. Each open track (b) is yours to define: the example directions below are hints to spark ideas, not requirements. Go further if you find something better.

Invertix Track

AI FOR RENEWABLE ENERGY OPERATIONS

a) DEDICATED CHALLENGE

Data-Center Siting & Power

WHERE TO BUILD, AND HOW TO POWER IT

AI is driving a wave of new data centres, and the binding constraint is increasingly electricity: is there grid capacity, what does power cost there, how clean is it, and can you sign a contract for renewables nearby? Siting a data centre is a multi-way trade-off between price, carbon, grid congestion, and connectivity. Can you help someone reason through where to build and how to power it?

Directions worth exploring:

- A tool that recommends locations for a given data-centre size and explains the trade-offs.
- A power-supply planner that mixes grid, PPA and on-site options against cost and carbon.
- A map that overlays grid capacity, prices, carbon and congestion to reveal good vs bad sites.

POINTERS	
European power grid	PyPSA-Eur prebuilt OSM network (Zenodo 18619025).
Power prices	Ember European electricity prices & costs.
Carbon & clean-power share	Ember Electricity Data Explorer.
Grid congestion	Ember Grids for data centres.
Existing data centres	OpenStreetMap.
Telco / fibre networks	ITU Transmission Networks (BBmaps) — connectivity proxy.
Useful assumptions	IEA Energy & AI (typical PUE and load factors).

b) OPEN TRACK — USING SATELLITE DATA FOR SOLAR

The open track is fully yours to define. Below are three example directions drawn from the resource pack to show what's possible. Pick one, combine ideas, or go somewhere entirely different — these are hints, not a menu.

Satellite Intelligence for Energy

MAPPING ENERGY ASSETS FROM SPACE

A new generation of geospatial foundation models turns satellite imagery into ready-to-use embeddings — one compact vector per 10-metre patch of the planet — so you can map things without training a vision model from scratch. Google DeepMind's AlphaEarth Foundations publishes annual global embeddings you can drop straight into a simple classifier. For energy that unlocks a lot: find every solar farm and rooftop array, spot wind turbines, map grid infrastructure, estimate rooftop solar potential, or flag vegetation creeping toward power lines.

Directions worth exploring:

- Detect and map solar PV (rooftop and utility-scale) across a region and estimate the installed capacity.
- Find wind turbines, or score land for wind and solar siting suitability.
- Map grid infrastructure, or flag wildfire and vegetation risk near transmission lines.
- Join embeddings with building energy-performance data to predict efficiency at scale.

This is one possible direction for the Invertix open track — a hint, not a requirement.

POINTERS

AlphaEarth embeddings	DeepMind AlphaEarth Foundations: 64-dim, 10 m, 2017–2025, hosted in Google Earth Engine (free account).
Raw imagery	Copernicus Data Space: free Sentinel-2 optical imagery if you want to work from pixels directly.
Labels & targets	OpenStreetMap (solar farms, substations, power lines) + Ember / IRENA capacity stats.
Reference repos	opengeos/leafmap · opengeos/geoai · Clay-foundation/model

Renewable Generation Forecasting

PREDICTING SOLAR AND WIND OUTPUT

Wind and solar output swings with the weather, and the whole grid has to be balanced against those swings in real time. Better short-term forecasts — from minutes to days ahead — cut balancing costs, reduce wasted renewable energy, and drive smarter trading. Can you build a model that predicts how much a site or region will generate, and show how much better it is than a naive guess?

Directions worth exploring:

- A day-ahead solar or wind forecast for a region from weather plus history, scored against a persistence baseline.
- A short-horizon nowcast (next 15–60 minutes) — the kind grid balancing actually needs.
- A forecast that feeds straight into a trading or flexibility decision.

This is one possible direction for the Invertix open track — a hint, not a requirement.

POINTERS

Weather data	Open-Meteo: free weather and forecast API (also on AWS Open Data).
PV modelling	pvlb-python: physics-based PV output from irradiance and panel parameters.
Actual generation	SMARD.de and ENTSO-E for realised wind/solar to train and score against.
Ready-made datasets	SOLETE (co-located wind, solar & meteorology) · Stanford sky-image PV dataset.
Reference repos	openclimatefix/open-data-pvnet · pvlb/pvlb-python · fmiddev/fmi-open-pv-forecast

Energy Trading Agents

FROM PRICES & NEWS TO SIGNALS

Electricity and gas prices move constantly with weather, demand, fuel costs and news. Traders turn all of that into positions. Can you build something that ingests prices (and maybe news sentiment), generates trading signals or a portfolio allocation, and honestly backtests whether the strategy would have worked?

Directions worth exploring:

- A signal generator (momentum, mean-reversion, fundamentals) with a clean backtest.
- A news-sentiment layer that nudges positions, with an agent that explains each trade.
- A multi-commodity allocation (power, gas) with explicit risk limits.

This is one possible direction for the Invertix open track — a hint, not a requirement.

POINTERS

US energy data & API	EIA Open Data: electricity, natural gas (Henry Hub), petroleum; free key, data back to 1990.
European prices	energy-charts.info (Fraunhofer ISE) · SMARD.de · Ember EU prices.
Load / generation / flows	ENTSO-E Transparency (free token by email).
News for sentiment	GDELT: large free global news dataset, filterable for energy.
Reference repos	albanpuech/energy-arbitrage-via-RL · AI4Finance-Foundation/FinRL · ekapope/EPEX-machine-learning

a) DEDICATED CHALLENGE

Digital Twin of a Solar Plant

BUILDING A LIVING MODEL OF REAL PV OPERATIONS

A digital twin is a model of a physical system that mirrors its real-time behaviour, enabling operators to detect faults, simulate what-ifs, and optimise performance without touching the physical plant. Enerparc is providing real monitoring data from two operating solar plants. Can you build a model — or a set of intelligent agents — that turns this raw data into actionable insight?

Directions worth exploring:

- Anomaly detection at the inverter or string level: flag underperformance relative to irradiation and expected output.
- Soiling detection: identify panels losing output due to dirt or shading using the Plant B dataset.
- Fault classification: cross-reference error codes with power drops to categorise incidents automatically.
- Performance ratio modelling: build a baseline model and quantify deviation per inverter over time.
- Service ticket intelligence: link monitoring anomalies to the historical ticket log to predict future failures.

POINTERS

Plant A dataset	Full monitoring data: per-inverter P_AC, irradiation, temperature, DC current/voltage, error codes, curtailment signals (EVU/DV), feed-in tariffs, service tickets.
Plant B dataset	Focused on soiling detection. Fewer data tracks. Fully optional — complete Plant A first.
File formats	main_monitoring_data.* in CSV, Parquet, and XLSB. Additional XLSX files for tariffs, system overview, tickets.
Error codes	errorcodes_description.xlsx — translation table for all inverter error codes.
System overview	System_Overview.xlsx — module type per inverter and installed capacity (kWp).

Dataset Information Slides — provided by Enerparc

Enerparc has prepared the following slides to explain the structure of the solar plants and the data tracks available in the dataset. Read these before diving into the files — they are the reference material for this challenge.

PV – Powerplant Structure

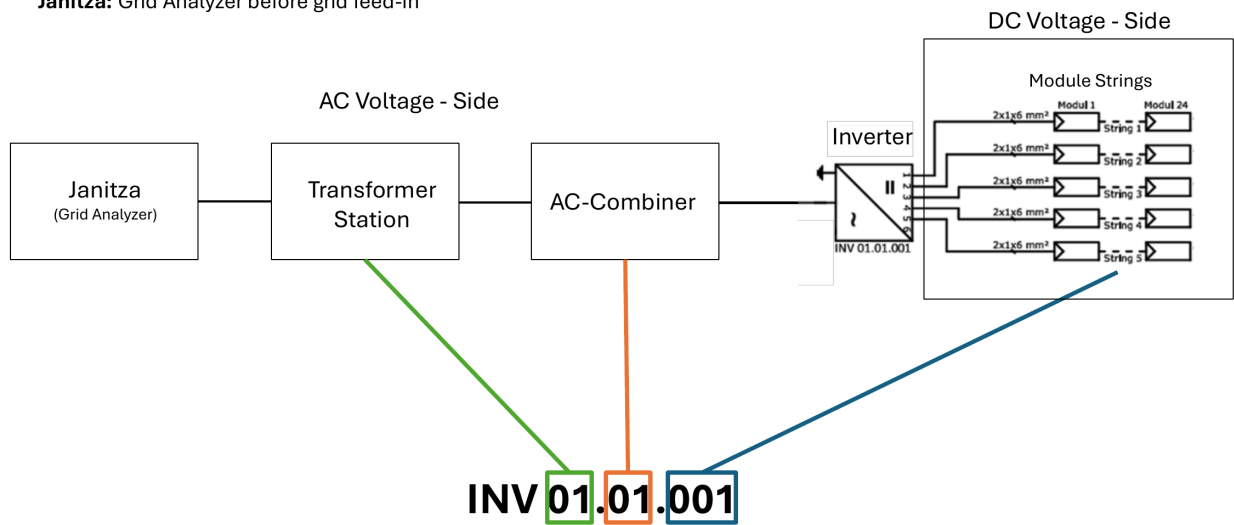
String: Multiple pv-modules connected in a row

Inverter: Converts DC-power from the several module strings to AC-power for grid feed-in

AC-Combiner: Combines multiple AC inverter outputs

Transformer station: Transforms AC-power to higher voltage

Janitza: Grid Analyzer before grid feed-in



Plant A · Slide 1 — PV powerplant structure: from module strings through inverter, AC-combiner and transformer station to grid feed-in via Janitza grid analyser.

Information about data tracks

Important data tracks:

- **Irradiation:** Measured by a pyranometer or a reference-cell-sensor on the plant. It is the most influential metric on plant power production
- **Altitude:** Calculated sun-height, helpful to determine the time of power production and filter timestamps without production
- **Inverter P_AC:** Inverter power output, this is the main power metric

Overview how these data tracks are named in folder “1. Main-monitoring-data”

INV 01.01.001 / P_AC (kW)
.....
INV 01.09.065 / P_AC (kW)
Plant / Altitude (°)
Plant / Irradiation_average (W/m²)

Plant A · Slide 2 — Essential data tracks: irradiation, sun altitude, and per-inverter P_AC (AC power output).

Information about data tracks

Useful data tracks:

- **Module/Ambient temperature:** Has a small influence on power production as solar cell efficiency varies with temperature
- **Inverter I_DC/U_DC:** Inverter input current and voltage. Can be used to determine inverter efficiency or voltage/current specific problems
- **EVU:** Plant curtailment by grid operator
- **DV:** Plant curtailment by plant operator (in case of e.g. negative electricity price)
 - Curtailment moments are undistinguishable from a plant outage without these datatracks
- **Errorcodes:** Errorcodes given by inverter software in case of inverter malfunction or other problems. Problem description are given in a separate table.
- **Feed in tariff:** electricity price given in eurocent / kWh
- **Janitza-Tracks:** Grid-input tracks
 - **Cos-Phi:** Ratio of active power/apparent power
 - **P_AC:** Total plant power that is fed into the grid

Overview how these data tracks are named in folder “1. Main-monitoring-data”

DRD11A / DV (%)
DRD11A / EVU (%)
INV 01.01.001 / I_DC_SUM (A)
INV 01.01.001 / U_DC (V)
.....
.....
INV 01.09.065 / I_DC_SUM (A)
INV 01.09.065 / U_DC (V)
Janitza UMG 604 - DRD11A / CosPhi_L1..L3
Janitza UMG 604 - DRD11A / I_AC_L1-L3 (A)
Janitza UMG 604 - DRD11A / P_AC_L1..L3 (kW)
Janitza UMG 604 - DRD11A / S_AC_L1..L3 (kVA)
Temperature Sensor / Ambient (°C)
Temperature Sensor / Module (°C)

Plant A · Slide 3 — Useful additional data tracks: temperature, DC current/voltage, curtailment signals (EVU/DV), error codes, feed-in tariff, and Janitza grid-quality tracks.

Overview Datatracks in folder “1. Main-monitoring-data”

- „main_monitoring_data.*“ as described before in CSV / Parquet or XLSB format. Content is identical for all file formats.

Overview folder “2. Additional Data”

- „feed-in-tariffs.xlsx“ contains monthly feed-in tariff
- „System_Overview.xlsx“ gives an overview which module type is connected to which inverter. Also installed capacity (kWp) per inverter can be found here.
- „Tickets.xlsx“ contains a selection of available service tickets in the sheets „2020-2026“ and „2019-2020“

Overview folder “3. Errorcodes”

- „errorcodes description.xlsx“ contains the translation table with meaning of each errorcode
- „errorcodes.*“ in CSV / Parquet or XLSB format. Content is identical for all file formats.

Plant A · Slide 4 — Folder and file structure overview for Plant A and Plant B datasets.

Plant B: Analysis is just focused on soiling detection. Therefore less data is provided. Please finish plant A first. This plant is fully optional and is totally fine just to work on plant A.

Overview Datatracks in folder “1. Main-monitoring-data”

- „main_monitoring_data.“ as described before in CSV / Parquet or XLSB format. Content is identical for all file formats.

Overview folder “2. Additional Data”

- “Coordinate.txt” contains the approx. location to calculate clear sky irradiation
- „feed-in-tariffs.xlsx“ contains monthly feed-in tariff
- „System_Overview.xlsx“ gives an overview which module type is connected to which inverter. Also installed capacity (kWp) per inverter can be found here.

Plant B · Slide 1 — Plant B is focused on soiling detection and provides a reduced set of data tracks. Optional: finish Plant A first.

b) OPEN TRACK — USING PV MONITORING & OPERATIONAL DATA

The Enerparc open track invites you to explore the full dataset freely. You are not limited to the digital twin framing — if you see a different problem worth solving with this data, go for it.

a) DEDICATED CHALLENGE

Grid Operation Agents

KEEPING A STRESSED GRID SECURE

A power grid has to survive the sudden loss of any single line or generator without anything else overloading — operators call this staying 'N-1 secure.' When a line trips, a human in a control room has minutes to react: open or close switches, turn some generators up and others down, adjust transformers, or curtail output. Could an AI agent reason through that situation and propose safe, low-cost actions — and explain them?

Directions worth exploring:

- An agent that takes an overloaded grid back to a safe state and narrates its reasoning step by step.
- Screening many 'what if this fails' scenarios quickly, then solving only the dangerous ones in detail.
- Comparing an LLM-driven approach against a simple rule-based or optimisation baseline.

POINTERS

Grid simulator	pandapower — Python power-flow & contingency engine with IEEE test grids built in.
Bigger / European grids	PyPSA-Eur prebuilt network from OpenStreetMap: Zenodo 18619025.
Pretrained grid model	Microsoft GridSFM_Open — small, CPU-runnable, fast grid-state prediction.
RL framing	Grid2Op and the L2RPN competitions — prior art on agents that operate grids.
Real load/generation data	SMARD.de (German TSO data) to make scenarios realistic.
Architecture blueprint	arXiv:2512.20789 (X-GridAgent) — LLM agent driving pandapower via MCP tools.

b) OPEN TRACK — USING GRID FOUNDATION MODELS

Grid Foundation Models

BUILDING ON A PRETRAINED GRID MODEL

'Foundation models' for power grids are an emerging idea: a model trained across many grids that can predict things like power flow or optimal dispatch far faster than a traditional solver. When prediction becomes near-instant, new things become possible — interactive what-if exploration, rapid screening, smarter optimisers. Can you build something genuinely useful on top of one, and stress-test where it holds up?

Directions worth exploring:

- An interactive 'drag a load, see the grid respond' explorer powered by a fast model.
- Using the model to rank risky failure cases, then checking the worst ones with a real solver.
- An honest evaluation: how accurate is it, and where does it break (grid size, unusual topologies)?

POINTERS

Pretrained model	Microsoft GridSFM_Open — small, CPU-runnable, weights downloadable. Best starting point.
Ground-truth solver	pandapower — generate reference answers to compare the model against.
Grids to test on	IEEE cases (in pandapower) and PyPSA-Eur Zenodo 18619025.
Reference repos	gridfm/gridfm-graphkit · gridfm/gridfm-datakit · AlexDeLos/GNN_OPF · arXiv:2406.07234

All suggested directions and pointers in this guide are starting points, not requirements. Verify data licences and API access yourself. Mentors are available for domain questions — ask early and often. Have fun.

Meta Quest 3 – Developer Setup Guide

Energy Hack

Note: The headsets are fully pre-configured – Developer Mode is active, accounts are set up, no login or reset required. You only need to **prepare your laptop**.

⚠ Do not leave the headset in direct sunlight. The lenses can focus sunlight onto the display and permanently burn the screen within seconds – even through a window.

What you need

- Your laptop (macOS, Windows, or Linux)
 - The provided **USB-C cable** (must support data transfer, not just charging)
 - ~10 minutes setup time
-

Step 1 – Install Android Platform Tools (ADB)

ADB (Android Debug Bridge) is the tool your laptop uses to communicate with the headset.

macOS

Option A – via Homebrew (recommended):

```
bash  
  
brew install android-platform-tools
```

If you don't have Homebrew yet:

```
bash  
  
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Option B – manual:

1. Download the [Android SDK Platform Tools](#) for macOS
2. Unzip to e.g. `~/platform-tools`
3. Add to your PATH:

```
bash
```

```
echo 'export PATH="$HOME/platform-tools:$PATH"' >> ~/.zshrc && source ~/.zshrc
```

Windows

Windows requires an additional USB driver on top of ADB.

1. Download the [Android SDK Platform Tools](#) for Windows and unzip it
 2. Add the folder to your system PATH (search "Edit environment variables" in the Start menu)
 3. Download and install the [Oculus ADB Driver](#) - this is required for the Quest to be recognized
 4. Reboot your machine
-

Linux

Ubuntu / Debian:

```
bash

sudo apt update && sudo apt install android-tools-adb
```

Arch:

```
bash

sudo pacman -S android-tools
```

Manual (any distro):

1. Download the [Android SDK Platform Tools](#) for Linux
2. Unzip and add to PATH:

```
bash

echo 'export PATH="$HOME/platform-tools:$PATH"' >> ~/.bashrc && source ~/.bashrc
```

Verify installation (all platforms):

```
bash

adb version
```

Step 2 - Connect the headset

1. Plug the **USB-C cable** into your laptop
2. **Put on the headset** - a popup will appear:
| *"Allow USB debugging?"*
3. Check **"Always allow from this computer"**
4. Tap **"Allow"**

| ⚠ No popup? Power cycle the headset and reconnect the cable.

Step 3 - Verify the connection

```
bash  
  
adb devices
```

Expected output:

```
List of devices attached  
1WMHHA123XXXXX  device
```

- Status `(device)` = you're connected, all good.
 - Status `(unauthorized)` = repeat Step 2 (confirm the popup in the headset).
 - No output = check your cable (must support data transfer, not just charging).
-

Step 4 - Deploy your app (install APK)

Once connected, push your app directly to the headset:

```
bash  
  
adb install my-app.apk
```

To update an already installed app:

```
bash  
  
adb install -r my-app.apk
```

Useful ADB commands

Command	Description
<code>adb devices</code>	List connected devices
<code>adb install app.apk</code>	Install an app
<code>adb install -r app.apk</code>	Update an existing app
<code>adb uninstall com.example.app</code>	Uninstall an app
<code>adb logcat</code>	Stream live logs from the headset
<code>adb shell</code>	Open a shell on the device

Troubleshooting

Issue	Fix
No popup in headset	Reconnect cable, restart headset
<code>unauthorized</code> in <code>adb devices</code>	Tap "Allow" in the headset
Cable not detected	Use a data transfer cable (not charge-only)
<code>adb: command not found</code>	Check installation, restart Terminal
Windows: device not recognized	Make sure the Oculus ADB Driver is installed and you rebooted

Come find us at the **info desk** – we'll help you right away. 🚀

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